

Shubham Singh

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Summary

Software engineer at Amazon with ~4 years of backend and platform work, including a shipped **agentic Q&A platform** with **hybrid retrieval** (BM25 + kNN) and **citation-grounded answers** that deflected ~90% of manual interpretation requests across 5 partner teams. Building production-style RAG and multi-agent infrastructure outside of work, contributing to LLM eval tooling (Giskard) and Microsoft's MarkItDown. Writes on production agent design at shusingh.github.io/writing.

Experience

Amazon

Aug 2022 – Present

Software Development Engineer II

Seattle, WA

- Built and shipped **MaverickIQ**, an agentic Q&A platform with **hybrid retrieval** (BM25 over control metadata fused with kNN over regulatory corpora), secure tool-calling on **AWS Bedrock + AgentCore + Strands SDK**, **citation-grounded responses**, and audit logging; deflected ~90% of manual interpretation requests across **5 partner teams**.
- Led cross-team design** of an internal compliance control-launch **developer platform** (React/TypeScript + AWS API Gateway/Lambda/DynamoDB); **sole author of the system HLD**, coordinating **3 partner teams across 3 timezones**; powers **2,000+ compliance control launches** and replaces multi-week engineering effort with self-serve hours.
- Designed a **stateful workflow orchestration platform** (Step Functions + DynamoDB) replacing brittle Lambda chains: per-shard Map parallelism, callbacks for long-running operations, lock-based resource ownership with stale-lock cleanup, and bounded compensation; cut workflow failure rate from **8% to <1%**.
- Designed a **PySpark compliance pipeline** processing **30M+ monthly records** with schema versioning, validation, and error isolation; reduced incident resolution time by **70%**.
- Owned production operations across **5 services** on rotating on-call; mentored a new-grad SDE to independent on-call readiness.

Open Source & Projects

AgentOps RAG (Python, FastAPI, OpenSearch, Redis, OTel) – production-style agentic RAG with explicit planner/retriever/critic/finalizer, JWT tenant isolation, async ingestion with DLQ replay, hybrid BM25 + vector retrieval, citation-grounded refusals, eval harness with citation precision/recall, and OTel traces across the full path. 2026

Giskard OSS Contribution (Python) – added JsonValid check with optional JSON Schema validation to Giskard's LLM eval library; iterated on maintainer feedback to validate schemas at instantiation via pydantic, handle unresolvable \$ref, and distinguish schema errors from validation failures. 2026

Strands Multi-Agent Orchestrator (Python, FastAPI, MCP) – planner, research, and data agents with MCP tools, SSE streaming, model-driven routing, mocked-LLM tests, and OTel tracing. 2026

Microsoft MarkItDown OSS Contribution (Python) – contributed RFC 822 / MIME .eml converter to Microsoft's document-to-Markdown LLM ingestion toolkit; extracts headers, body, attachment metadata, with vector-test coverage. 2026

Technical Skills

AI Systems: Multi-Agent Orchestration, RAG, Hybrid Retrieval (BM25+kNN), Tool-Calling, MCP, Citation Grounding, LLM-as-Judge, HITL

AI Infra: AWS Bedrock, AgentCore, Strands SDK, Eval Harnesses, Prompt/Model Versioning, OpenTelemetry

Languages: Python, Go, TypeScript, Java, SQL, C/C++

Systems: Step Functions, Lambda, DynamoDB, OpenSearch, Postgres, Redis, S3, API Gateway, Docker, Kubernetes, CI/CD, IaC

Backend/Data: PySpark, ETL Pipelines, FastAPI, REST APIs, Microservices, Event-Driven Architecture

Education

Indiana University Bloomington

M.S. in Data Science

Jan 2021 – Jul 2022

Bloomington, IN